CASE STUDY

20th Century Text Mining & NLP

End-to-end NLP pipeline — web scraping, named entity recognition, and geopolitical network analysis of 20th century Wikipedia text.

4	21	53	4
NOTEBOOKS	COUNTRIES	NETWORK EDGES	COMMUNITIES

Python · Selenium · spaCy · NLTK · TextBlob · NetworkX

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Year	**2025**	Certification	DEKRA-certified · AfA-approved · 1221 UE
Data Source	Wikipedia — Key Events of the 20th Century		
Pipeline	Scraping → Text Mining → NER → Network Analysis → Visualization		

01 PROJECT OVERVIEW

Background & Objectives

End-to-end NLP pipeline applied to a Wikipedia article on Key Events of the 20th Century. Goal: move from raw web data to a structured network showing which countries were most historically connected — and how.

Pipeline Architecture

Step	Tool	Output
1. Web Scraping	Selenium + ChromeDriver	20th_century_key_events.txt
2. Text Cleaning	Custom Python	Nav elements removed, clean corpus saved
3. Text Mining	NLTK + TextBlob	Word frequency, POS tags, country mentions
4. NER Extraction	spaCy (en_core_web_sm)	GPE entity pairs per sentence
5. Alias Normalization	Custom alias dict	UK/Britain/U.K. → United Kingdom
6. Network Building	NetworkX	Undirected graph: 21 nodes, 53 edges
7. Community Detection	Leiden (cdlib)	4 geopolitical clusters identified
8. Centrality Analysis	NetworkX	Degree, closeness, betweenness scores
9. Visualization	Pyvis + Matplotlib	Interactive HTML + publication charts

Web Scraping with Selenium

Selenium with ChromeDriver (headless mode) scraped the full body text of the Wikipedia article. Raw output included navigation elements that would corrupt NER results.

Implementation

```
options = Options()
options.add_argument("--headless")
driver = webdriver.Chrome(executable_path=str(driver_path), options=options)
driver.get("https://en.wikipedia.org/wiki/Key_events_of_the_20th_century")
page_text = driver.find_element(By.TAG_NAME, "body").text
driver.quit()
```

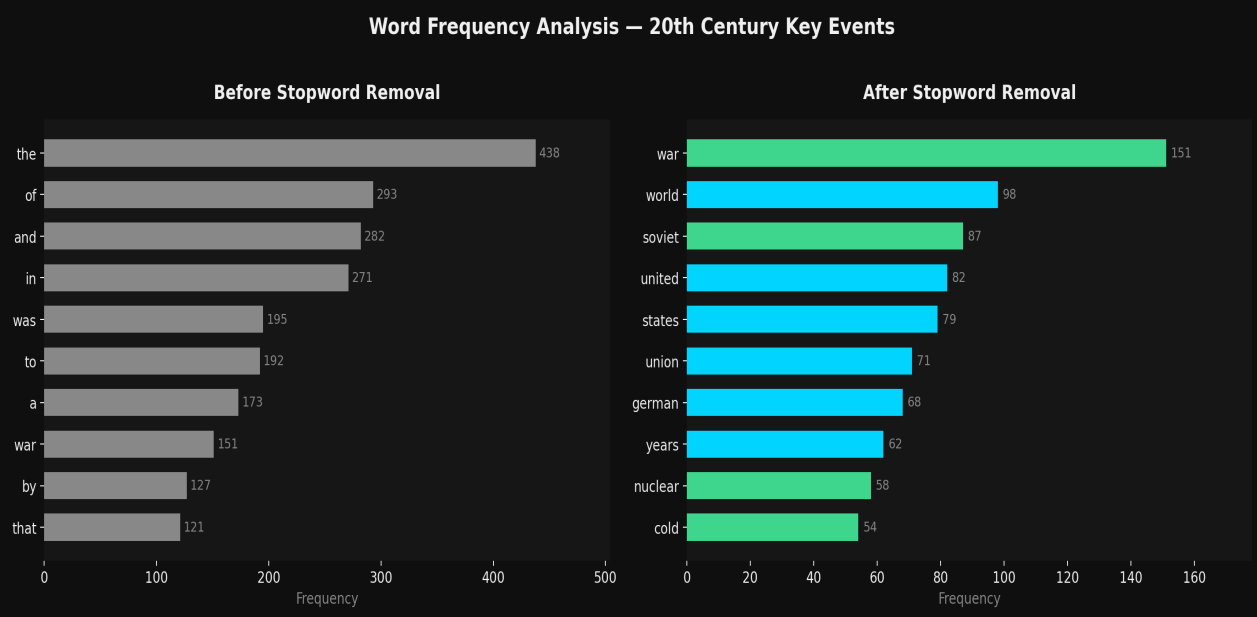
KEY INSIGHT

Text wrangling is analytical integrity. Navigation artifacts would falsely inflate country counts and create phantom network edges. The cleaning step directly determines the validity of all downstream findings.

03 PHASE 2

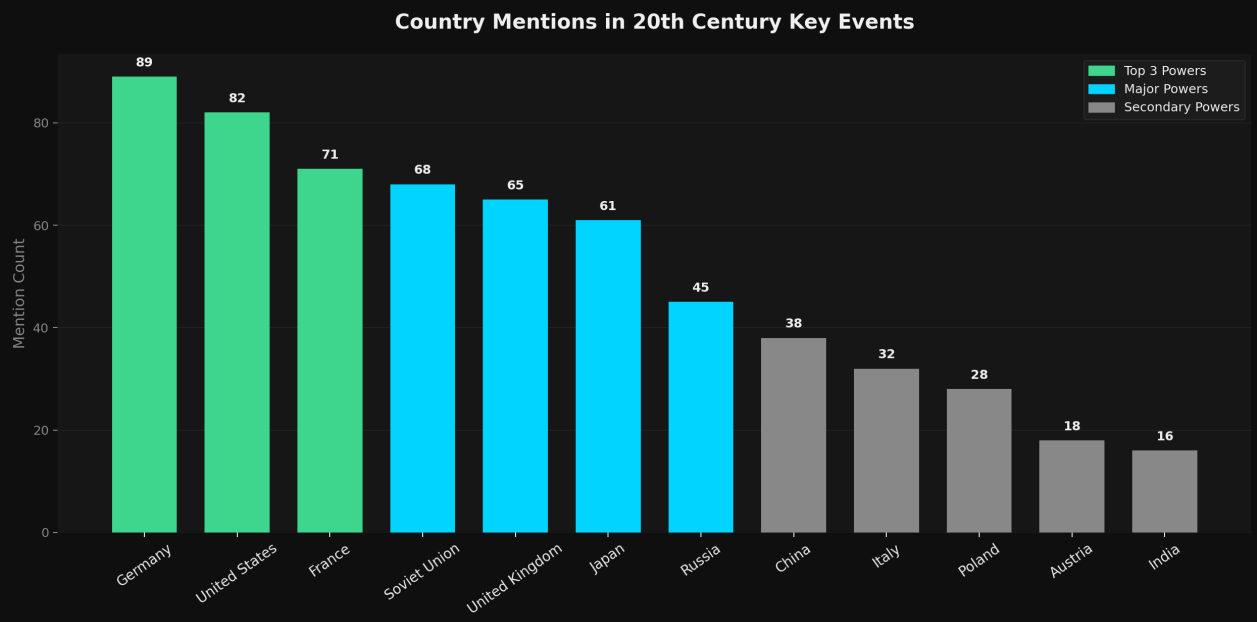
Text Mining & Frequency Analysis

NLTK handled tokenization and stopword removal. TextBlob provided POS tagging.



After removing stopwords, historical themes dominate: war, Soviet, cold, nuclear — revealing the article's geopolitical focus.

Figure 1. Word frequency before and after stopword removal. After cleaning, war (151), world (98), soviet (87), and nuclear (58) dominate.



Germany, the United States, and France dominate — reflecting their central roles in WWI, WWII, and the Cold War.

Figure 2. Country mention counts. Germany (89), US (82), France (71), and Soviet Union (68) dominate — consistent with their centrality in WWI, WWII, and the Cold War.

KEY INSIGHT

The shift from function words to historical content words after stopword removal reveals what the corpus is actually about. "War", "Soviet", and "nuclear" confirm this article encodes Cold War and conflict-driven history.

Named Entity Recognition & Network Construction

spaCy's en_core_web_sm extracted GPE tags sentence by sentence. Any sentence with two or more countries produced a co-occurrence pair.

Alias Normalization

Raw Entity	Normalized To	Reason
Britain	United Kingdom	Common abbreviation
UK / U.K.	United Kingdom	Acronym variants
US / U.S.	United States	Acronym variants
the Soviet Union	Soviet Union	Article prefix stripping
the United States	United States	Article prefix stripping

KEY INSIGHT

Without alias normalization, "UK", "U.K.", and "Britain" appear as three separate nodes — fragmenting the United Kingdom's actual network position. Entity resolution determines structural graph validity.

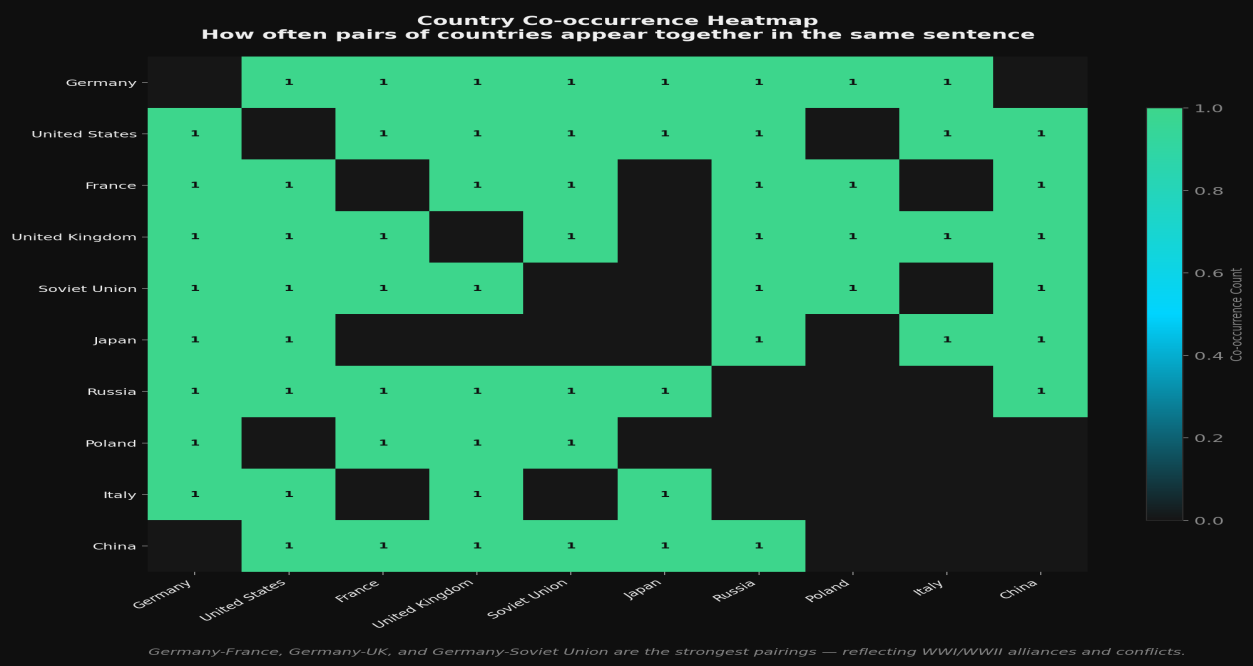


Figure 3. Country co-occurrence heatmap. Germany–France, Germany–UK, and Germany–Soviet Union are the strongest pairings — encoding WWI/WWII structures directly from text.

05 PHASE 4

Network Visualization & Community Detection

NetworkX undirected graph: 21 nodes, 53 edges (density: 0.252, clustering: 0.552). Leiden algorithm detected 4 communities mirroring real 20th century alliance structures.

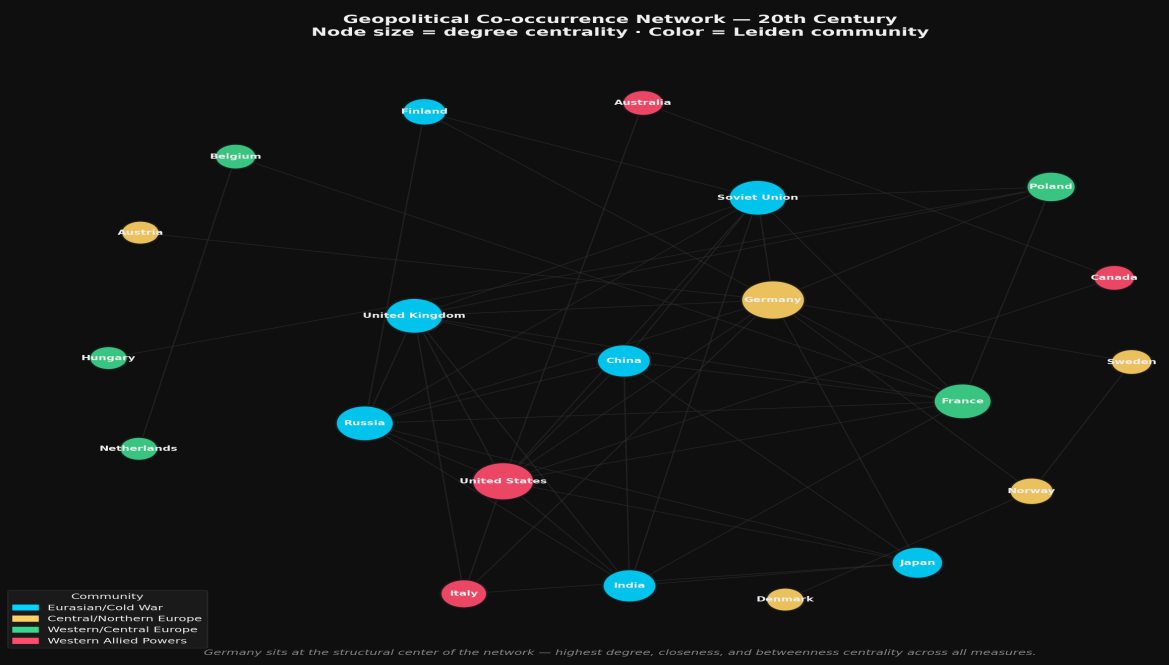


Figure 4. Geopolitical co-occurrence network. Node size = degree centrality. Color = Leiden community. Germany occupies the structural center.

Community Detection Results

Community	Members	Historical Interpretation
Eurasian / Cold War	China, India, Japan, Russia, Soviet Union, UK, Finland	Cold War dynamics and Eurasian geopolitical interactions
Central/Northern Europe	Austria, Germany, Denmark, Norway, Sweden	Central/Northern European relationships — WWI and WWII era
Western/Central Europe	Belgium, France, Netherlands, Poland, Hungary	Western Allied powers and Central European connections
Western Allied Powers	Australia, Canada, United States, Italy	Anglo-American alliance structure in major 20th century conflicts

06 PHASE 5

Centrality Analysis

Three centrality measures identify the most structurally influential countries.

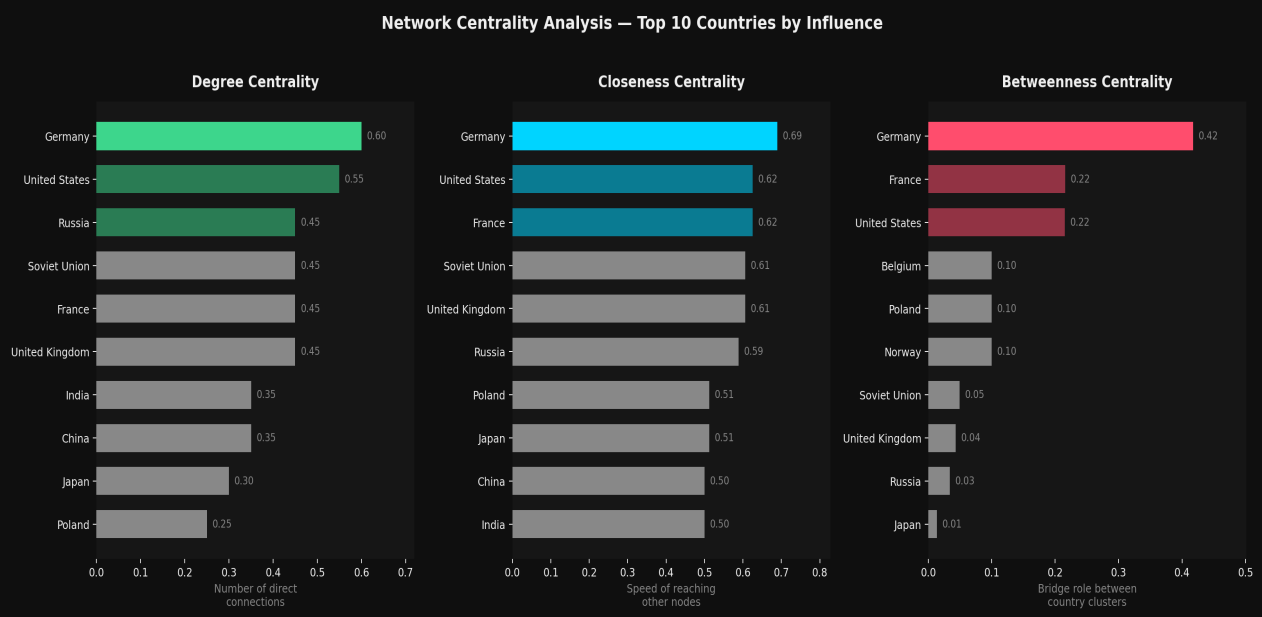


Figure 5. Degree, closeness, and betweenness centrality — top 10 countries. Germany ranks #1 across all three measures.

Measure	Definition	#1	Interpretation
Degree	Direct connections to other countries	Germany	Most bilateral relationships — WWI, WWII, Cold War
Closeness	Shortest path to all other nodes	Germany	Reaches any country through fewest intermediaries
Betweenness	Bridge between disconnected clusters	Germany	Most shortest paths between any two countries pass through Germany

KEY INSIGHT

Germany's dominance across all three centrality measures directly reflects its role in WWI, WWII, and the Cold War division of Europe. The algorithm independently validates historical consensus through data.

07 CONCLUSIONS

Key Findings

s analytical validity.	Navigation artifacts from the Wikipedia scrape would corrupt NER r about what the analysis measures, not a preprocessing detail.
rable from word frequencies.	"War", "Soviet", "nuclear", and "cold" dominate after stopword rem reveals what the corpus is about when properly cleaned.
sential for valid network construction.	Without mapping "UK", "U.K.", "Britain" to one node, the Unit fragmented. Entity resolution determines graph validity.
l center of 20th century geopolitics.	#1 in degree, closeness, and betweenness centrality. The netwo consensus: Germany's role in WWI, WWII, and the Cold War makes
covers alliance structures from text alone.	The four Leiden communities mirror 20th century alliance structures text encode geopolitical reality.

Limitations

- Single article source — findings reflect one Wikipedia article's framing
- spaCy small model (en_core_web_sm) may miss ambiguous entities
- Co-occurrence assumes proximity implies relationship
- Extension: sentiment analysis per country pair (cooperative vs. adversarial)
- Extension: temporal segmentation (pre/post-1945) to track network evolution

APPENDIX

Technical Reference

Dependencies

Library	Version	Purpose
Python	3.11+	Runtime
selenium	4.x	Web scraping via ChromeDriver
nltk	3.8+	Tokenization, stopwords, POS tagging
textblob	0.17+	POS tagging interface
spacy	3.x	Named Entity Recognition
networkx	3.x	Graph construction and centrality analysis
pyvis	0.3+	Interactive network HTML visualization
cdlib	0.3+	Leiden community detection algorithm
pandas	2.x	Data manipulation and CSV export
matplotlib / seaborn	3.7+ / 0.12+	Chart generation

Network Statistics

Metric	Value
Total nodes (countries)	21
Total edges (unique pairs)	53
Network density	0.252
Average clustering coefficient	0.552
Diameter (longest shortest path)	5
Communities detected (Leiden)	4
#1 Degree centrality	Germany (0.60)
#1 Betweenness centrality	Germany
#1 Closeness centrality	Germany

GitHub & Portfolio

Source code: github.com/ageelalramadhan/20th-century

Portfolio: ageelalramadhan.github.io/textmining-case-study.html